

Math 514, F13
Exam 1

Name: *Answer*
SSN: xxx-xx-

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.

Problem	Points	Score
1	15	
2	15	
3	20	
4	20	
5	30	
Total	100	

Good Luck !

Problem 1. (15 points) Two dice are rolled. Let A be the event that the scores on the two dice are the same. Let B be the event that the total of the scores is five or less. Find $P(A)$, $P(B)$, and $P(A|B)$.

Solution: $A = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\}$

$$\Rightarrow P(A) = \frac{6}{36} = \frac{1}{6}$$

$$B = \left\{ \underbrace{(1, 1)}_2, \underbrace{(1, 2), (2, 1)}_3, \underbrace{(1, 3), (2, 2), (3, 1)}_4, \right. \\ \left. (1, 4), (2, 3), (3, 2), (4, 1) \right\}$$

$$\Rightarrow P(B) = \frac{10}{36} = \frac{5}{18}$$

$$A \cap B = \{(1, 1), (2, 2)\} \Rightarrow P(A \cap B) = \frac{2}{36} = \frac{1}{18}$$

$$\text{thus } P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{1}{18}}{\frac{5}{18}} = \frac{1}{5}$$

Problem 2. (15 points) Ten people attend a meeting. Two of them do not own a car, four of them each own one car, three of them each own two cars, and one of them owns four cars. Let X be the number of cars owned by a randomly chosen person attending the meeting. Find $E[X]$, $E[X^2]$, and $\text{Var}(X)$.

Solution: $E[X] = 0 \cdot \frac{2}{10} + 1 \cdot \frac{4}{10} + 2 \cdot \frac{3}{10} + 4 \cdot \frac{1}{10}$
 $= \frac{4}{10} + \frac{6}{10} + \frac{4}{10} = \frac{14}{10} = \frac{7}{5}$

$$E[X^2] = 0^2 \cdot \frac{2}{10} + 1^2 \cdot \frac{4}{10} + 2^2 \cdot \frac{3}{10} + 4^2 \cdot \frac{1}{10}$$

 $= \frac{4}{10} + \frac{12}{10} + \frac{16}{10}$
 $= \frac{32}{10} = \frac{16}{5}$

$$\text{Var}(X) = E[X^2] - E[X]^2$$

 $= \frac{16}{5} - \left(\frac{7}{5}\right)^2$
 $= \frac{31}{25}$

Problem 3. (20 points) In a school, 70% of students have access to the internet at home. Let X be the number of students who have the internet access at home in a group of 6 students chosen at random. Find $E(X)$, $\text{Var}(X)$, and $P(X \geq 4)$.

Solution: Let $X_i = \begin{cases} 1 & \text{if the } i\text{-th student has internet access} \\ 0 & \text{otherwise} \end{cases}$

$$E[X_i] = 1 \times 0.7 + 0 \times 0 = 0.7$$

$$\text{Var}(X_i) = E[X_i^2] - E[X_i]^2 = 0.7 - 0.7^2 = 0.21$$

It holds $X = \sum_{i=1}^6 X_i$, thus

$$E[X] = \sum_{i=1}^6 E[X_i] = 6 \times 0.7 = 4.2$$

$$\text{Var}(X) = \sum_{i=1}^6 \text{Var}(X_i) = 6 \times 0.21 = 1.26$$

$$\begin{aligned} P(X \geq 4) &= P(X=4) + P(X=5) + P(X=6) \\ &= \binom{6}{2} 0.7^4 \times 0.3^2 + \binom{6}{1} 0.7^5 \times 0.3 + \binom{6}{0} 0.7^6 \\ &\approx 0.7443 \end{aligned}$$

Problem 4. (20 points) Assume that Tom wins \$1 at probability $1/4$, \$2 at $1/4$ and loses \$1 at $1/2$ at a certain bet. Each bet is independent. What is the probability that Tom will win at most \$20 in total after 100 bets?

Solution: Let X_i denote the amount that Tom win at i -th bet.

$$E[X_i] = 1 \cdot \frac{1}{4} + 2 \cdot \frac{1}{4} + (-1) \cdot \frac{1}{2} = \frac{1}{4}$$

$$\begin{aligned} \text{Var}(X_i) &= E[X_i^2] - E[X_i]^2 = 1 \cdot \frac{1}{4} + 2^2 \cdot \frac{1}{4} + (-1)^2 \cdot \frac{1}{2} - \left(\frac{1}{4}\right)^2 \\ &= \frac{27}{16} \end{aligned}$$

$$\begin{aligned} \text{Let } X = \sum_{i=1}^{100} X_i, \text{ then } E[X] &= 100 \cdot \frac{1}{4} = 25 \\ \text{Var}(X) &= 100 \cdot \frac{27}{16} = \frac{2700}{16} \end{aligned}$$

$$\begin{aligned} P(X \leq 20) &= P\left(Z \leq \frac{20-25}{\sqrt{\frac{2700}{16}}}\right) = P\left(Z \leq \frac{-5}{\frac{3}{2}\sqrt{27}}\right) \\ &= P\left(Z \leq -\frac{2}{\sqrt{27}}\right) = P\left(Z > \frac{2}{\sqrt{27}}\right) \\ &= 1 - \Phi\left(\frac{2}{\sqrt{27}}\right) \approx 1 - \Phi(0.3849) \\ &\approx 0.3502. \end{aligned}$$

Problem 5. (30 points) The price of a certain stock $S(t)$, $t \geq 0$ is modeled by a geometric Brownian motion process with the *weekly* drift parameter $\mu = 0.0006$ and volatility parameter $\sigma = 0.035$.

(a) Find the probability that the price of the stock will be up at the end of each of the next 2 weeks.

$$\begin{aligned} \text{Solution: } P\left(\frac{S(1)}{S(0)} > 1\right) &= P\left(\log\left(\frac{S(1)}{S(0)}\right) > 0\right) = P(X(1) > 0) \\ &\stackrel{\substack{\downarrow \\ S(1) > S(0)}}{=} P\left(Z > \frac{0 - \mu \cdot 1}{\sqrt{\sigma^2 \cdot 1}}\right) = P\left(Z > \frac{0 - 0.0006}{0.035}\right) \\ &= P(Z > -0.0171) = P(Z < 0.0171) \\ &= \Phi(0.0171) \approx 0.5068 \end{aligned}$$

$$\begin{aligned} \Rightarrow P(S(2) > S(1) > S(0)) &= P(S(2) > S(1)) \cdot P(S(1) > S(0)) \\ &= 0.5068^2 \\ &\approx 0.2569 \end{aligned}$$

(b) Estimate the probability that the price of the stock will be up by at least 5% at the end of 4 weeks.

$$\begin{aligned} \text{Solution: } P(S(4) > 1.05 S(0)) &= P\left(\frac{S(4)}{S(0)} > 1.05\right) \\ &= P\left(\log\left(\frac{S(4)}{S(0)}\right) > \log 1.05\right) \approx P(X(4) > 0.0488) \\ &= P\left(Z > \frac{0.0488 - \mu \cdot 4}{\sqrt{\sigma^2 \cdot 4}}\right) = P\left(Z > \frac{0.0488 - 0.0024}{0.07}\right) \\ &\approx P(Z > 0.6629) = 1 - P(Z < 0.6629) \\ &= 1 - \Phi(0.6629) \\ &= 0.2537 \end{aligned}$$

(c) What is the expected stock price at the end of one year if the initial price $S(0) = 100$.

$$\begin{aligned} \text{Solution: } E[S(52)] &= S(0) e^{\mu \cdot 52 + \frac{\sigma^2}{2} \cdot 52} \xrightarrow{52 \text{ weeks}} \\ &= 100 \cdot e^{(0.0006) \cdot 52 + \frac{(0.035)^2}{2} \cdot 52} \\ &\approx 100 \cdot e^{0.063} \\ &\approx 106.51 \end{aligned}$$